Fiscal Year:	FY 2010	**Task Last Updated:**	FY 07/22/2010
PI Name:	Niesel, David William Ph.D.		
Project Title:	Streptococcus pneumoniae Gene Expression and Virulence Potential in the Space Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77555-1019	**Congressional District:**	14
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	02/01/2000	**End Date:**	01/31/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-118/ISS NOTE: Received extension to 1/31/2010 (previously 1/31/2009) per A. Chu (2/2009) NOTE: Received extension to 1/31/2009 per A. Chu/ARC (6/2008) 6/23/06--End date is 7/31/2008, per A. Chu (ARC) renewed. Updated Aug. 26, 2005.		
Key Personnel Changes/Previous PI:	Nina Williams - Research assistant / Graduate Student ; Zinzi Blell, Graduate student; Audrie Medina, postbac student, Stacy Agar,PhD - postdoc.		
COI Name (Institution):			
Grant/Contract No.:	NCC2-1160		
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Task Description:	The long term focus and goal for the NASA and the US space program is extended space flight and the long-term habitation of space environments including the moon, Mars and beyond. While we have learned much during extensive human visits to low earth orbit about the effects of microgravity and the space environment on human physiology, it is clear that extended space residence will represent a challenge to crew members health. In this regard, much evidence has accumulated about the effects of this unique environment on many aspects of human physiology. This includes profound alterations in the immune function which may compromise element of our host defense. This combined with the recent observed changes in the physiology, gene expression or the virulence potential of microbes that cause infectious disease and prevention of infectious disease in space travelers becomes a major concern. In our studies, we are investigating, Streptococcus pneumoniae, a human commensal bacterium that with dysfunctional host defense can become an opportunistic pathogen. S. pneumoniae is an important human pathogen and the leading cause of bacterial pneumonia, meningitis, and otitis media in the very old or young or those with compromised immune systems. Importantly, this microbe has been isolated from the nasal pharynx of crew members before flight. Our long term goal is to understand the effect of microgravity and the space environment on the ability of this opportunistic microbe and others to produce disease and to devise mitigation strategies for their control. Our hypothesis for these studies is that S. pneumoniae will alter gene and protein expression and its virulence potential in low shear modeled microgravity (LSMMG) and in the space environment. Experimentation to address this hypothesis has been organized into three specific objectives. Experiments will be performed to identify and characterize S. pneumoniae genes and proteins which are differentially expressed in response to the space environment or in response to LSMMG. The second objective will be to investigate the expression of known S. pneumoniae virulence genes or properties in LSMMG. Finally, we will determine if S. pneumoniae genes differentially expressed in LSMMG or during space flight are also expressed during animal infection. Overall, our results will provide new information about a significant human pathogen, add to our knowledge the S. pneumoniae pathogenic mechanism and also provide basic information on bacterial model system of gene and protein expression in the space environment. Importantly, these studies will also provide for a rationale risk assessment for this organism and other opportunistic microbes in the context of long-term space flight and habitation and contribute to our understanding of the virulence of this bacterium on earth. This experimentation will advance NASA objectives, and provide much needed information to advance human health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study will reveal new information about S. pneumoniae, a medically important opportunistic bacterial pathogen. The data collected will advance our understanding of the pathogenic mechanism of the pneumococcus and how it adapts to different environments. Understanding how this microbe responds to the space environment will additionally allow a rationale risk assessment for this and other opportunistic bacterial pathogens. This could be important in the success of long-term space missions and for the habitation of space. These space flight results will benefit NASA's goals of protecting human health during long duration space travel and potentially provide information that could lead to development of countermeasures - to detect illnesses, protect individuals from infectious diseases or treat these diseases in space. Additionally, new information from these studies on the S. pneumoniae pathogenic mechanism will have direct earth benefits in terms of understanding the diseases mediated by this pathogen and could lead to novel vaccines and therapeutics for this bacterial pathogen.
Task Progress:	Since our last report we have had the opportunity to include the SPEGIS experiment on the STS 118 shuttle flight. For the SPEGIS experiment, new hardware was developed at NASA Ames and designed to accommodate a shuttle mid-deck and ISS experimental protocol. The hardware utilized commercial culture tubes with hardware and the culture tubes having O-ring seals which provided three levels of containment of the bacterial culture. Optimal culture time was optimized at 15.5 hrs. Viability and stowage requirements for S. pneumoniae during launch phase, cold stowage prior to the initiation of growth, growth parameters were designed to provide sufficient bacterial numbers for analysis, and the optimal storage for sample return to earth. The experiment used three canisters, which contains culture tubes with different bacterial concentrations to allow recovery of mid-logarithmic and late logarithmic phase cells. Since bacterial cells will be frozen after growth, we focused on global transcriptional profiling using DNA microarrays, and assessing the total proteome between ground control and flight conditions. S. pneumoniae 6304 was flown in SPEGIS hardware on STS 118. Mid-logarithmic bacterial cultures were launched and maintained at 40C in the MERLIN incubator. After transfer to the International Space Station (ISS), growth was at 370C for 15.5 hrs. Parallel cultures prepared with the flight samples were incubated in the flight simulator facility in the Space Life Science Laboratory (SLS) at KSC. Cultures were frozen in the MELFI nitrogen vapor freezer and returned to earth in a passive cooler at -320C. Cultures were frozen at -800C upon landing. Bacterial numbers were determined by serial dilution and plating. Both cultures reached stationary phase by optical density measurement. However, bacterial viability was ~5 logs lower than expected which likely resulted from the freezing and transport of the cultures after growth, but were consistent with late logarithmic / stationary phase cultures. Interestingly, two colonial variants were recovered in different proportions from the flight and ground control cultures. Electron microscopy performed on these colonial variants showed no differences in cellular or capsule size or ultrastructure. Enzyme activities for neuraminidase and ß-galactosidase from flight cultures were 1.4- and ~1.4- fold higher than that observed for ground controls respectively. We also examined the levels of the potent S. pneumoniae, pore-forming toxin, pneumolysin and found a 3-fold higher level of hemolytic activity in flight cultures versus ground control cultures. These enzymatic activities are associated with adherence and binding to host cells. Pneumolysin is a pore-forming toxin associated with cellular and tissue damage during systemic infection. These data show that changes are observed in virulence properties of S. pneumoniae following flight. Two dimensional gel electrophoresis was used to compare protein expression between the flight-grown and ground control cultures and 181 proteins were detected. Comparison of the protein profiles showed 17 differentially expressed S. pneumoniae proteins between flight-grown and ground controls. Of these 17 proteins, 13 were up-regulated and 4 down-regulated. Protein spots were picked and subjected to MS sequencing. Up-regulated proteins included phosphoglycerate kinase and galactokinase, which are proteins that participate in metabolism and are associated with pneumococcal virulence. Proteins expressed at reduced levels include the CLP protease and G3PDH. The Clp protease is a regulatory protein that has been linked to the virulence activity of the pneumococcus. Global transcriptional profiling was carried out on total RNA isolated from frozen flight and ground control samples by using Streptococcus pneumoniae Genome Microarrays. Four spots for each gene were included on each chip and four

	biological replicates for flight and ground control samples (16 replicates per gene) were analyzed. ANOVA analysis of the data from four biological replicates resulted in the identification of 104 differentially expressed transcripts. Moreover, the 56 differentially expressed genes were shown to be up regulated in flight samples, while 48 were up regulated in ground samples. The gene nomenclature and functional annotation provided by The Institute for Genomic Research for S. pneumoniae TIGR4 (TIGR; http://cmr.tigr.org/) was used to identified different functional groups. Differentially expressed genes belonged to almost all functional groups and mapped across the S. pneumoniae genome. More than 1/3 of these genes were identified as hypothetical or unknown genes. The next three most observed groups were those in the “Other” group followed by the cell envelope, and transport and binding. These were equally split between those that were up- and down-regulated in flight. Genes encoding regulatory functions were the next most observed group and were unique among the functional groups, in that the majority of the genes were shown to be up-regulated in flight. Hierarchical clustering revealed two blocks of genes; one block which was up regulated in flight and the other block was up regulated in the ground samples. There were 32 virulence genes of S. pneumoniae that were differentially regulated including genes associated with the initiation of infection, production of systemic disease, and the stress response. Changes in virulence and gene and protein expression profiles under flight conditions may indicate the potential of the bacterium to initiate infection or produce disease in these environments. We believe that this would have significant implications for long-term space flight or space habitation. One manuscript has been published showing alterations in global transcriptional activity under modeled microgravity. A second describing alterations in the virulence activity and total proteome analysis under LSMMG was submitted and is under re-vision. A third manuscript with the bulk of the results from STS 118 has been drafted. These data will be combined with results from STS 123 for submission to Nature Communications.
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